

SOLUTION QUIZ-01 BCS-5D

QUESTION-01

- 1) C
- 2) B
- 3) B

TRUE/FALSE

- i. False
- ii. False

QUESTION-02

a) Calculate the Transmission Delay on Link 1:

$$\text{Transmission Delay on Link 1} = \frac{\text{Packet Size}}{\text{Bandwidth}}$$

$$T_{D1} = \frac{32 \times 10^3 \text{ bits}}{10 \times 10^6 \text{ bps}} = \frac{32}{10^4} \text{ seconds} = 3.2 \text{ ms}$$

b) Calculate the Transmission Delay on Link 3:

$$\text{Transmission Delay on Link 3} = \frac{\text{Packet Size}}{\text{Bandwidth}}$$

$$T_{D3} = \frac{32 \times 10^3 \text{ bits}}{5 \times 10^6 \text{ bps}} = \frac{32}{5 \times 10^3} \text{ seconds} = 6.4 \text{ ms}$$

c) Determine the Required Bandwidth X on Link 2:

The total end-to-end delay is the sum of all transmission delays and propagation delays:

$$\text{Total End-to-End Delay} = (T_{D1} + T_{D2} + T_{D3}) + (P_{D1} + P_{D2} + P_{D3})$$

$$50 \text{ ms} = (3.2 \text{ ms} + T_{D2} + 6.4 \text{ ms}) + (4 \text{ ms} + 6 \text{ ms} + 10 \text{ ms})$$

First, sum the propagation delays:

$$P_{total} = 4 \text{ ms} + 6 \text{ ms} + 10 \text{ ms} = 20 \text{ ms}$$

Now, substitute into the equation:

$$50 \text{ ms} = (3.2 \text{ ms} + T_{D2} + 6.4 \text{ ms}) + 20 \text{ ms}$$

Subtract the propagation delays and the transmission delays for Link 1 and Link 3:

$$50 \text{ ms} - 20 \text{ ms} - 3.2 \text{ ms} - 6.4 \text{ ms} = T_{D2}$$

$$T_{D2} = 20.4 \text{ ms}$$

Finally, calculate the required bandwidth X on Link 2 using the transmission delay T_{D2} :

$$T_{D2} = \frac{\text{Packet Size}}{X}$$

$$20.4 \text{ ms} = \frac{32 \times 10^3 \text{ bits}}{X \times 10^6 \text{ bps}}$$

Solve for X :

$$X = \frac{32 \times 10^3 \text{ bits}}{20.4 \times 10^{-3} \text{ seconds}} = \frac{32 \times 10^3}{20.4 \times 10^{-3}} \text{ bps}$$

$$X = \frac{32 \times 10^3}{0.0204} \text{ Mbps} \approx 1.57 \text{ Mbps}$$